

P12. Uncertainty-Aware Safety-Critical Decision and Control for Autonomous Vehicles at Unsignalized Intersections

中英双语博士精读笔记

Ran Yu; Zhuoren Li; Lu Xiong; Wei Han; Bo Leng

2025 · arXiv preprint

Theme / 主题: Uncertainty-aware safe RL + HOCBF

Method family / 方法族: Uncertainty-aware RL + high-order CBF / 不确定性感知强化学习与高阶控制屏障函数

PDF pages / 页数: 7 **Relevance / 相关度:** 84/100

Source / 来源: <https://arxiv.org/abs/2505.19939>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“Uncertainty-aware safe RL + HOCBF”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Uncertainty-aware safe RL + HOCBF. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Single-agent urban autonomous-driving framing rather than centralized conflict-zone scheduling.

1.2 核心科学问题 / Core Scientific Question

如何让 RL 在无信号交叉口中既保持灵活决策，又在不确定性升高时由 HOCBF 强制保证安全？

How can RL remain flexible while a high-order CBF safety filter intervenes under uncertainty at unsignalized intersections?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Risk-aware ensemble distributional RL estimates policy uncertainty, then a high-order control barrier function acts as a safety filter.

- 安全/避碰贡献 (safety contribution): HOCBF safety filter dynamically adjusts constraints using joint uncertainty in unsignalized-intersection tasks.
- 平滑/停止避免贡献 (smoothing contribution): Balances safety and traffic efficiency by switching between flexible RL behavior and conservative HOCBF intervention.
- 创新力度评判 (reviewer judgement): 较强: 把 ensemble distributional RL 的不确定性与 HOCBF 过滤结合, 适合作为学习调度器安全外壳。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Uncertainty-aware RL + high-order CBF	不确定性感知强化学习与高阶控制屏障函数	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: HOCBF 安全过滤 / High-order CBF safety filter, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

Uncertainty-Aware Safety-Critical Decision and Control for Autonomous Vehicles at Unsignalized Intersections

Ran Yu, Zhuoren Li, Lu Xiong, Wei Han, Bo Leng

Abstract—Reinforcement learning (RL) has demonstrated potential in autonomous driving (AD) decision tasks. However, applying RL to urban AD, particularly in intersection scenarios, still faces significant challenges. The lack of safety constraints makes RL vulnerable to risks. Additionally, cognitive limitations and environmental randomness can lead to unreliable decisions in safety-critical scenarios. Therefore, it is essential to quantify confidence in RL decisions to improve safety. This paper proposes an Uncertainty-aware Safety-Critical Decision and Control (USDC) framework, which generates a risk-averse policy by constructing a risk-aware ensemble distributional RL, while estimating uncertainty to quantify the policy's reliability. Subsequently, a high-order control barrier function (HOCBF) is employed as a safety filter to minimize intervention policy while dynamically enhancing constraints based on uncertainty. The ensemble critics evaluate both HOCBF and RL policies, embedding uncertainty to achieve dynamic switching between safe and flexible strategies, thereby balancing safety and efficiency. Simulation tests on unsignalized intersections in multiple tasks indicate that USDC can improve safety while maintaining traffic efficiency compared to baselines.

I. INTRODUCTION

Autonomous driving (AD) is becoming a significant focus of global innovation due to its potential in energy saving, emission reduction, traffic efficiency, road safety, and releasing drivers [1]. Although rule-based autonomous vehicles (AV) are effective in reducing accident rates in straightforward, merge, and exit scenarios, the accident frequency in turning scenarios is 1.98 times higher than that of human-driven vehicles (HDVs) [2]. Due to the limitations of predefined rules in covering all driving scenarios, AVs struggle to make flexible decisions when faced with multiple oncoming HDVs, potentially causing traffic congestion or rear-end collisions [3], [4].

Recently, reinforcement learning (RL) has demonstrated remarkable proficiency in decision-making tasks, such as those on the highway [5], merge [6], and intersection [7], due to its ability to optimize high-dimensional state spaces through continuous interaction with dynamic environments. However, even well-trained RL agents encounter considerable challenges in ensuring the safety of policies due to the lack of safety constraints [8]. Consequently, Safe RL has emerged as a paradigm for RL applications, with the

objective of maximizing the expected cumulative reward while simultaneously satisfying safety constraints [9].

A typical Safe RL architecture is based on the Constrained Markov Decision Process (CMDP) framework, which guarantees safety by constraining the cumulative expected cost below a threshold. This is usually addressed by solving the constrained optimization problem using Lagrangian multiplier methods [10], [11] or trust region methods [12], [13]. However, as the policy gradually converges near the safety threshold, learning a safe policy becomes more challenging due to the sparsity of the cost signal. Another common paradigm for Safe RL involves performing safety corrections to the RL policy, such as using action masks [14] and safe energy functions [15]–[17]. Action masks identify unsafe actions by predefining a safe action set and limiting unsafe actions from participating in the network's updates. Nevertheless, such methods are typically applicable only to discrete action spaces. Control barrier functions (CBF) are a significant representative of safety energy functions and are widely regarded as a well-established method in safety control [18]. However, many CBF approaches require the design of intricate prior functions, which limits their adaptability and generalization in complex dynamic scenarios. Furthermore, the existence of high relative degree between the standard CBF and the underlying system control inputs requires a more conservative barrier function for safety, otherwise the safety can be compromised due to the inability to constrain the higher-order dynamics resulting in transient transgressions of the lower-order constraints. Learning-based safe energy functions have garnered significant attention because of their independence from prior knowledge. However, they are deficient in clearly defined and interpretable safety guarantee mechanisms.

RL decisions typically yield only outcomes without revealing the uncertainty. This inherent ambiguity can pose safety risks in AD tasks [19]. This is particularly evident in scenarios involving long-tailed problems or out-of-distribution (OOD) events, where the reliability of RL policies is significantly compromised. The quantification of the confidence level of RL decisions facilitates the identification of RL actions that pose potential safety risks due to high uncertainty. This, in turn, enables the reduction or avoidance of dangerous decisions in OOD scenarios. Uncertainty can be categorized into two main types: epistemic uncertainty (EU) and aleatoric uncertainty (AU) [20]. EU reflects data-dependent model or parameter uncertainty, which arises from lack of knowledge and can be reduced by observing more data; AU is irreducible, reflecting the stochastic nature of

This work has been submitted to the IEEE for possible publication. Copyright may be transferred without notice, after which this version may no longer be accessible. (corresponding author: Zhuoren Li).

Ran Yu, Zhuoren Li, Lu Xiong, Wei Han and Bo Leng are with the School of Automotive Studies, Tongji University, Shanghai 201804, China (email: ranyu@tongji.edu.cn; 1911055@tongji.edu.cn; xiong.lu@tongji.edu.cn; tjhanwei@tongji.edu.cn; lengbo@tongji.edu.cn).

图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

environmental dynamics [21]. Recent works in the field of AD have demonstrated that uncertainty-aware RL has garnered significant attention and focus. A common approach is to use a backup policy to replace the RL policy when the estimated uncertainty exceeds a threshold or falls outside a safety set [8], [22]–[25]. However, the set or threshold is often limited to specific scenarios. Furthermore, some backup strategies, such as deceleration or braking, may negatively impact traffic efficiency.

To this end, in order to enhance the ability of RL to handle safety-critical and OOD scenarios while achieving a balance between safety and traffic efficiency, we propose the Uncertainty-aware Safety-critical Decision and Control (USDC) framework, as illustrated in Fig. 1. This framework leverages deep ensemble (DE) [26] to estimate the joint uncertainty (JU) that encompasses both EU and AU. By dynamically adjusting safety constraints based on JU, it effectively balances safety and efficiency. The main contributions are as follows:

- We propose a risk-aware distributional RL ensemble architecture that combines the original and Fixed Prior Networks (FPN) to build the critic. It quantifies tail risk in the reward distribution and generates a risk-averse policy, while jointly estimating EU and AU for dual uncertainty-driven decision-making.
- A high-order control barrier function (HOCBF) is constructed, which ensures safety using only relative distance. Moreover, HOCBF incorporates JU to dynamically adjust safety constraints, balancing safety and traffic efficiency.
- Based on the JU distribution, the HOCBF and RL policies are evaluated, ensuring that the RL policy performs no worse than the HOCBF under low uncertainty and favors safer policies under high uncertainty.

II. PRELIMINARIES

A. Distributional Reinforcement Learning

RL can be modeled as a Markov Decision Process (MDP) defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \rho_0, \gamma)$. $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the transition probability function, $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function, $\rho_0 : \mathcal{S} \rightarrow [0, 1]$ is the initial state distribution, and γ is the discount factor for future reward. The policy $\pi : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{A})$ is a map from given states to a probability distribution on action space. Standard MDP aims at maximizing the agent's cumulative discounted reward $\mathcal{L}(\pi) = \mathbb{E}_{\pi, \mathcal{P}} [\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$. For $\pi \in \Pi$, the Q function $Q^\pi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ can be defined as $Q^\pi(s, a) := \mathbb{E}_{\pi, \mathcal{P}} [\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$, which satisfies the following Bellman equation:

$$Q(s, a) := \mathbb{E}[r(s, a)] + \gamma \mathbb{E}_{\pi, \mathcal{P}}[Q(s', a')]. \quad (1)$$

In the distributional RL setup, the distribution over returns $Z(s, a)$ is estimated, and its expectation is maximized to obtain $Q(s, a) = \mathbb{E}[Z(s, a)]$. Hence, we can rewrite (1) as:

$$Z(s, a) \stackrel{D}{=} r(s, a) + \gamma Z(s', a'), \quad (2)$$

where $\stackrel{D}{=}$ indicates that both sides of the equation are distributed according to the same distribution. Let $F_Z(z) = \mathbb{P}(Z \leq z)$ denote the cumulative distribution function (CDF) of random variable Z , the quantile function F_Z^{-1} can be represented as the inverse of CDF. Given quantile fraction τ , we have $F_Z^{-1} := \inf\{z \in \mathbb{R} : \tau \leq F_Z(z)\}$.

Following [27], we defined a series of quantile fractions $\{\tau_i\}_{i=0, \dots, N}$, $\hat{\tau}_i = (\tau_i + \tau_{i+1})/2$, and the pairwise temporal difference (TD) error between two quantile fractions $\hat{\tau}_i, \hat{\tau}_j$:

$$\delta_{ij} = r + \gamma \left[Z_{\hat{\tau}_i}^\theta(s', a') - \alpha \log \pi_{\bar{\phi}}(a'|s') \right] - Z_{\hat{\tau}_j}^\theta(s, a) \quad (3)$$

where $\bar{\theta}, \bar{\theta}, \bar{\phi}$ are parameters of target critic, critic, and target actor network, respectively. α is the temperature parameter. We then train the critic $Z_{\hat{\tau}_i}^\theta(s, a)$ using quantile regression by minimizing the weighted pairwise Huber loss [28] with threshold κ :

$$\mathcal{L}_Z(\theta) = \frac{1}{|\mathcal{B}|} \sum_{(s, a, r, s') \in \mathcal{B}} \sum_{i=0}^{N_q-1} \sum_{j=0}^{N_q-1} (\tau_{i+1} - \tau_i) \rho_{\hat{\tau}_j}^\kappa(\delta_{ij}) \quad (4)$$

$$\rho_{\hat{\tau}_j}^\kappa(\delta_{ij}) = |\tau - \mathbb{I}\{\delta_{ij} < 0\}| \frac{\mathcal{L}_\kappa(\delta_{ij})}{\kappa}, \quad (5)$$

$$\mathcal{L}_\kappa(\delta_{ij}) = \begin{cases} \frac{1}{2} \delta_{ij}^2, & \text{if } |\delta_{ij}| \leq \kappa \\ \kappa (|\delta_{ij}| - \frac{1}{2} \kappa), & \text{otherwise,} \end{cases} \quad (6)$$

where $\mathcal{B}$ is a minibatch of transitions sampled from a replay buffer, N_q denotes the number of quantile points over $[0, 1]$. Then, the objective of actor is to maximize the Q -return:

$$L_\pi(\phi) = \mathbb{E}_{s_t \sim \mathcal{D}, a_t \sim \pi_\phi} [\alpha \log(\pi_\phi(a_t|s_t)) - Q_\theta(s_t, a_t)], \quad (7)$$

where $Q_\theta(s, a) = \sum_{i=0}^{N_q} (\tau_{i+1} - \tau_i) Z_{\hat{\tau}_i}^\theta(s, a)$, $\mathcal{D}$ is the transitions replay buffer.

B. High-Order Control Barrier Functions

Consider an input-affine control system given by:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}, \quad (8)$$

where the system state $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^n$ and the control input $\mathbf{u} \in \mathcal{U} \subset \mathbb{R}^{u_n}$. f and g are locally Lipschitz.

Definition 1 (Forward invariant set): The set $\mathcal{C}$ is forward invariant for system (8) if for every initial condition $\mathbf{x}(t_0) \in \mathcal{C}$, $\mathbf{x}(t) \in \mathcal{C}$ for $\forall t \geq t_0$. For a continuously differentiable function $h : \mathcal{X} \rightarrow \mathbb{R}$, let

$$\mathcal{C} = \{\mathbf{x} \in \mathcal{X} : h(\mathbf{x}) \geq 0\}. \quad (9)$$

Definition 2 (CBF [18]): Denote class $\mathcal{K}$ function is a function $\alpha : [0, a) \rightarrow [0, \infty)$, $a > 0$ that is strictly increasing and with $\alpha(0) = 0$. Given the superlevel set $\mathcal{C}$ as in (9), h is a CBF if there exists a class $\mathcal{K}$ function α such that

$$\sup_{\mathbf{u} \in \mathcal{U}} [L_f h(\mathbf{x}) + L_g h(\mathbf{x})\mathbf{u}] \geq -\alpha(h(\mathbf{x})), \forall \mathbf{x} \in \mathcal{C}, \quad (10)$$

where $L_f h, L_g h$ denote the Lie derivatives along f and g , respectively. In the discrete system perspective, we consider a discrete-time system $\mathbf{x}_{k+1} = f(\mathbf{x}_k) + g(\mathbf{x}_k)\mathbf{u}_k$, where $k \in \mathbb{N}$ denotes the time step. Then, given a set $\mathcal{C}$ as in (9), continuous function h is a candidate discrete CBF if there exists a class $\mathcal{K}$ function satisfying $\alpha(x) \leq x$ such that:

$$\sup_{\mathbf{u} \in \mathcal{U}} [\Delta h(\mathbf{x}, \mathbf{u}) + \alpha(h(\mathbf{x}))] \geq 0, \forall \mathbf{x} \in \mathcal{C}, \quad (11)$$

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

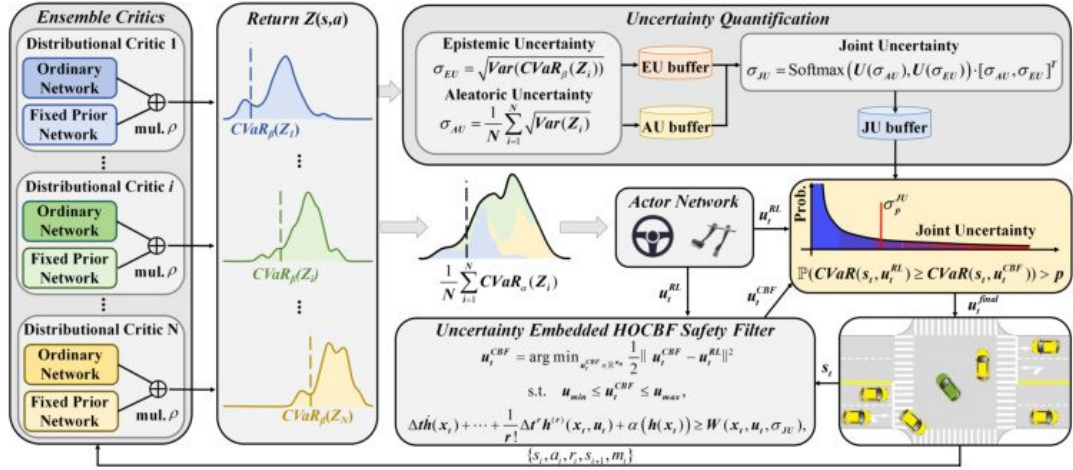


Fig. 1. Diagram of our framework.

图 3: 核心方法或算法页 / Core method or algorithm page (p.3, figure_crop). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

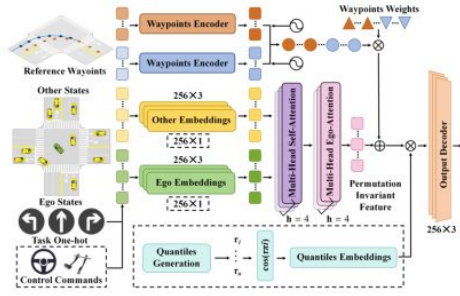


Fig. 3. Details of designed network. The components within the dashed box are used only in the critic network.

preclude the use of more advanced techniques for quantile generation.

D. Experiment Results

We compare USDC with the following baselines: distributional soft actor critic (DSAC [27]); pure-DSAC, which uses a similarly structured multi-layer perceptron (MLP) network for the network part; USDC without using HOCBF as a safety filter; and Dir-CBF, which directly applies HOCBF filtering to each action generated by a well-trained USDC agent without considering uncertainty. During training, each algorithm is trained three times with different random seeds. In each episode, tasks with randomly generated EVs are assigned, including left turn (LT), go straight (GS), and right turn (GT). During the testing phase, each algorithm is evaluated on 200 episodes for each of the three tasks.

The learning curves in Fig. 4 and the test results in Table I indicate that the USDC with a probabilistic safety filter improves performance while balancing traffic efficiency and safety. Despite an increase in FR compared to USDC without HOCBF, integrating the critic to estimate the $CVaR$ of u^{RL} and u^{CBF} provides safety guarantees under high uncertainty, resulting in a 11.0% reduction in CR compared to DSAC. In low uncertainty scenarios, the policy that maximizes $CVaR$ is selected by comprehensively assessing the long-term returns and instantaneous risks of u^{RL} and u^{CBF} . This results in final performance metrics of 207.3 for AER and 7.77 m/s for AEV, matching or even surpassing other baselines. It is noteworthy that, although USDC increases FR by 3.5% compared to the USDC without HOCBF, its dynamic policy switching achieved through uncertainty quantification effectively avoids the drawbacks caused by over-conservatism in direct CBF. As shown in Table I, although the direct CBF method controls CR at 2.8%, its FR is as high as 27.8%. Additionally, the SR, AER, and AEV are reduced by 19.9%, 49.3%, and 12.6% respectively compared to USDC, indicating that relying solely on safety constraints may compromise the balance between efficiency and safety for AV.

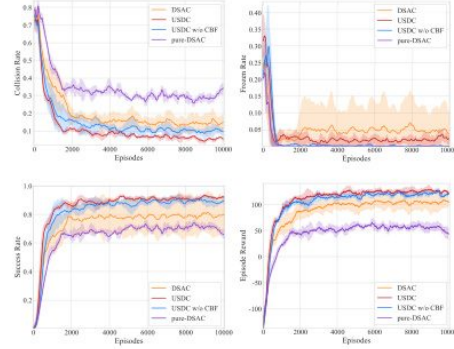


Fig. 4. Training curves of experiment. Solid lines correspond to the mean and the shaded regions correspond to 95% confidence interval over 3 runs.

TABLE I
COMPARE PERFORMANCE ON THREE DRIVING TASKS.

Tasks	Algorithms	SR(%)	FR(%)	CR(%)	AER	AEV(m/s)
LT	Pure-DSAC	56.5	0.5	43.0	-22.7±48.2	7.99±0.64
	DSAC	82.5	1.0	19.5	100.9±47.5	7.00±1.33
	Dir-CBF	72.0	28.0	4.5	6.1±64.0	6.34±1.51
	USDC (ours)	91.0	5.0	6.0	129.9±47.1	7.71±0.44
	USDC w/o CBF	86.5	0.5	15.5	117.4±46.8	7.91±0.45
GS	Pure-DSAC	62.5	3.5	34.0	30.8±52.4	7.85±0.74
	DSAC	73.0	7.5	23.0	147.2±53.4	7.73±0.53
	Dir-CBF	56.0	48.0	3.0	58.8±69.5	6.81±1.04
	USDC (ours)	88.5	9.0	5.0	182.7±47.7	7.39±0.37
	USDC w/o CBF	83.5	3.0	14.5	172.4±49.5	7.57±0.43
RT	Pure-DSAC	88.5	0.5	11.5	190.8±30.1	7.98±0.57
	DSAC	98.0	0.5	1.5	267.8±19.5	8.06±0.42
	Dir-CBF	92.0	7.5	1.0	232.0±37.1	7.21±0.87
	USDC (ours)	100.0	0.0	0.0	309.2±12.1	8.01±0.22
	USDC w/o CBF	98.5	0.0	1.5	278.9±17.1	7.99±0.29
MEAN	Pure-DSAC	69.2	1.5	30.2	66.3±43.6	7.94±0.65
	DSAC	84.5	3.0	14.7	172.0±40.1	7.60±0.76
	Dir-CBF	73.3	27.8	2.8	99.0±56.8	6.79±1.14
	USDC (ours)	93.2	4.7	3.7	207.3±35.6	7.77±0.34
	USDC w/o CBF	89.5	1.2	10.5	189.6±37.8	7.82±0.39

¹ SR, FR, CR, AER, and AEV stand for success rate, frozen rate, collision rate, average episode reward, and average episode velocity, respectively. The definitions are similar to those in [40], except that FR is defined as a runtime exceeding 20 seconds (200 steps).

² Bold: best performance. Policy update frequency $f_{\pi} = 10$ Hz.

V. CONCLUSIONS

In this paper, we propose an uncertainty-aware safety-critical decision and control framework. USDC generates a risk-averse policy by constructing a risk-aware distributional ensemble RL, while estimating uncertainty to quantify the reliability of the policy. FPN and Bootstrapping are employed to enhance the diversity of critics when encountering safety-critical scenarios. Subsequently, a HOCBF is employed as a safety filter to minimize intervention policy, while dynamically enhancing constraints based on uncertainty. Instead of relying on a specific threshold, the ensemble critics evaluate both HOCBF and RL policies simultaneously based on the

图 4: 实验或结果页 / Experiments or results page (p.6, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

JU distribution. This ensures that the RL policy performs no worse than the HOCBF under low uncertainty and favors safer policies under high uncertainty. Experimental results for left turn, right turn, and go straight driving tasks demonstrate that USDC improves safety compared to baseline methods while maintaining traffic efficiency. The forward invariance of the HOCBF may be violated because of input constraints. Additionally, the HOCBF in this paper only guarantees safety for a single step and does not account for the effects of changes in obstacle states. Future work can be extended to a more robust safety filter to improve overall safety.

REFERENCES

- [1] H. Wang, W. Shao, C. Sun, K. Yang, D. Cao, and J. Li, "A survey on an emerging safety challenge for autonomous vehicles: Safety of the intended functionality," *Engineering*, vol. 33, pp. 17–34, 2024.
- [2] M. Abdel-Aty and S. Ding, "A matched case-control analysis of autonomous vs human-driven vehicle accidents," *Nat. Commun.*, vol. 15, no. 1, p. 4931, 2024.
- [3] S. Noh, "Decision-making framework for autonomous driving at road intersections: Safeguarding against collision, overly conservative behavior, and violation vehicles," *IEEE Trans. Ind. Electron.*, vol. 66, no. 4, pp. 3275–3286, 2019.
- [4] M. T. Ashraf, K. Dey, S. Mishra, and M. T. Rahman, "Extracting rules from autonomous vehicle-involved crashes by applying decision tree and association rule methods," *Transp. Res. Rec.*, vol. 2675, no. 11, pp. 522 – 533, 2021.
- [5] Z. Li, L. Xiong, B. Leng, P. Xu, and Z. Fu, "Safe reinforcement learning of lane change decision making with risk-fused constraint," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2023, pp. 1313–1319.
- [6] G. Li, W. Zhou, S. Lin, S. Li, and X. Qu, "On-ramp merging for highway autonomous driving: An application of a new safety indicator in deep reinforcement learning," *Automot. Innov.*, vol. 6, no. 3, pp. 453 – 465, 2023.
- [7] H. Hu, D. Chu, J. Yin, and L. Lu, "Double deep q-networks based game-theoretic equilibrium control of automated vehicles at autonomous intersection," *Automot. Innov.*, vol. 7, no. 4, pp. 571 – 587, 2024.
- [8] K. Yang, S. Li, Y. Chen, D. Cao, and X. Tang, "Towards safe decision-making for autonomous vehicles at unsignalized intersections," *IEEE Trans. Veh. Technol.*, vol. 74, no. 3, pp. 3830–3842, 2025.
- [9] S. Gu, L. Yang, Y. Du, G. Chen, F. Walter, J. Wang, and A. Knoll, "A review of safe reinforcement learning: Methods, theories, and applications," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11 216–11 235, 2024.
- [10] A. Stooke, J. Achiam, and P. Abbeel, "Responsive safety in reinforcement learning by pid lagrangian methods," in *Int. Conf. Mach. Learn. PMLR*, 2020, pp. 9133–9143.
- [11] H. Honari, A. M. S. Enayati, M. G. Tamizi, and H. Najjaran, "Meta sac-lag: Towards deployable safe reinforcement learning via metagradient-based hyperparameter tuning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, IEEE, 2024, pp. 619–626.
- [12] J. Achiam, D. Held, A. Tamar, and P. Abbeel, "Constrained policy optimization," in *Int. Conf. Mach. Learn.*, PMLR, 2017, pp. 22–31.
- [13] Y. Zhang, Q. Vuong, and K. Ross, "First order constrained optimization in policy space," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 15 338–15 349, 2020.
- [14] Z. Shixin, P. Feng, J. Anni, Z. Hao, and G. Qiuqi, "The unmanned vehicle on-ramp merging model based on am-mappo algorithm," *Sci. Rep.*, vol. 14, no. 1, p. 19416, 2024.
- [15] H. Ma, C. Liu, S. E. Li, S. Zheng, W. Sun, and J. Chen, "Learn zero-constraint-violation safe policy in model-free constrained reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 2327–2341, 2025.
- [16] H. Zheng, H. Ma, S. Zheng, S. E. Li, and J. Wang, "Synthesize efficient safety certificates for learning-based safe control using magnitude regularization," in *IEEE Int. Conf. Robot. Autom.*, 2024, pp. 545–551.
- [17] X. Wang, "Ensuring safety of learning-based motion planners using control barrier functions," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 4773–4780, 2022.
- [18] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *Eur. Control Conf. (ECC)*, 2019, pp. 3420–3431.
- [19] W. Zhou, Z. Cao, N. Deng, K. Jiang, and D. Yang, "Identify, estimate and bound the uncertainty of reinforcement learning for autonomous driving," *IEEE Trans. Intell. Transport. Syst.*, vol. 24, no. 8, pp. 7932–7942, 2023.
- [20] E. Hüllermeier and W. Waegeman, "Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods," *Mach. Learn.*, vol. 110, no. 3, pp. 457–506, 2021.
- [21] S. Wang and S. Wen, "Safe control against uncertainty: A comprehensive review of control barrier function strategies," *IEEE Syst. Man Cybern. Mag.*, vol. 11, no. 1, pp. 34–47, 2025.
- [22] C.-J. Hoel, K. Wolff, and L. Laine, "Ensemble quantile networks: Uncertainty-aware reinforcement learning with applications in autonomous driving," *IEEE Trans. Intell. Transport. Syst.*, vol. 24, no. 6, pp. 6030–6041, 2023.
- [23] Z. Zhang, Q. Liu, Y. Li, K. Lin, and L. Li, "Safe reinforcement learning in autonomous driving with epistemic uncertainty estimation," *IEEE Trans. Intell. Transport. Syst.*, vol. 25, no. 10, pp. 13 653–13 666, 2024.
- [24] K. Yang, X. Tang, S. Qiu, S. Jin, Z. Wei, and H. Wang, "Towards robust decision-making for autonomous driving on highway," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11 251–11 263, 2023.
- [25] X. Tang, G. Zhong, S. Li, K. Yang, K. Shu, D. Cao, and X. Lin, "Uncertainty-aware decision-making for autonomous driving at uncontrolled intersections," *IEEE Trans. Intell. Transport. Syst.*, vol. 24, no. 9, pp. 9725–9735, 2023.
- [26] M. A. Ganaie, M. Hu, A. K. Malik, M. Tanveer, and P. N. Suganthan, "Ensemble deep learning: A review," *Eng. Appl. Artif. Intell.*, vol. 115, p. 105151, 2022.
- [27] X. Ma, L. Xia, Z. Zhou, J. Yang, and Q. Zhao, "Dsac: Distributional soft actor critic for risk-sensitive reinforcement learning," *arXiv preprint arXiv:2004.14547*, 2020.
- [28] P. J. Huber, "Robust estimation of a location parameter," in *Breakthroughs in statistics: Methodology and distribution*. Springer, 1992, pp. 492–518.
- [29] W. Xiao and C. Belta, "High-order control barrier functions," *IEEE Trans. Autom. Control*, vol. 67, no. 7, pp. 3655–3662, 2022.
- [30] Y. Xiong, D.-H. Zhai, M. Tavakoli, and Y. Xia, "Discrete-time control barrier function: High-order case and adaptive case," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3231–3239, 2023.
- [31] R. Rockafellar and S. Uryasev, "Conditional value-at-risk for general loss distributions," *J. Bank. Financ.*, vol. 26, no. 7, pp. 1443–1471, 2002.
- [32] A. Rame and M. Cord, "Dice: Diversity in deep ensembles via conditional redundancy adversarial estimation," in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [33] I. Osband, C. Blundell, A. Pritzel, and B. Van Roy, "Deep exploration via bootstrapped dqn," *Adv. Neural Inf. Process. Syst.*, vol. 29, 2016.
- [34] I. Osband, J. Aslanides, and A. Cassirer, "Randomized prior functions for deep reinforcement learning," *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.
- [35] R. Rajamani, *Vehicle dynamics and control*. Springer Science & Business Media, 2011.
- [36] J. Xu and B. Alrifaei, "High-order control barrier functions: Insights and a truncated taylor-based formulation," *arXiv preprint arXiv:2503.15014*, 2025.
- [37] J. A. E. Andersson, J. Gillis, G. Horn, J. B. Rawlings, and M. Diehl, "CasADi – A software framework for nonlinear optimization and optimal control," *Mathematical Programming Computation*, 2018.
- [38] E. Leurent, "An environment for autonomous driving decision-making," <https://github.com/eleurent/highway-env>, 2018.
- [39] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E*, pp. 1805–1824, Jul 2002.
- [40] B. Leng, R. Yu, W. Han, L. Xiong, Z. Li, and H. Huang, "Risk-aware reinforcement learning for autonomous driving: Improving safety when driving through intersection," *arXiv preprint arXiv:2503.19690*, 2025.

图 5: 补充精读页 / Supplemental close-reading page (p.7, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Uncertainty-aware safe RL + HOCBF：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 II. P RELIMINARIES

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.3 III. M ETHODOLOGIES

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.5 IV. I MPLEMENTATIONS

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.6 2 Bold: best performance. Policy update frequency $f = 10$ Hz.

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.6 V. CONCLUSIONS

章节目标 / Learning goal: 提炼作者承认的边界，并把它转化为博士课题的可拓展问题。

Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument: 结论不只是总结结果，更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus: 把 future work 与前文假设逐一对应，寻找可发表的补强点。

审稿追问 / Reviewer question: 如果我是审稿人，我会要求补哪一个实验、定理或消融？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：如何让 RL 在无信号交叉口中既保持灵活决策，又在不确定性升高时由 HOCBF 强制保证安全？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：安全函数可正确表达交叉口危险集合。；车辆动力学和相对状态可观测。；不确定性估计与真实风险有足够相关性。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“HOCBF 安全过滤 / High-order CBF safety filter”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Risk-aware ensemble distributional RL estimates policy uncertainty, then a high-order control barrier function acts as a safety filter., 输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Simulation tests on multiple unsignalized-intersection tasks against safety and efficiency baselines.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	HOCBF 安全过滤 / High-order CBF safety filter
3	核心求解 / Core solver	Risk-aware ensemble distributional RL estimates policy uncertainty, then a high-order control barrier function acts as a safety filter.
4	安全接口 / Safety interface	HOCBF safety filter dynamically adjusts constraints using joint uncertainty in unsignalized-intersection tasks.
5	平滑与效率 / Smoothing and efficiency	Balances safety and traffic efficiency by switching between flexible RL behavior and conservative HOCBF intervention.
6	验证 / Validation	Simulation tests on multiple unsignalized-intersection tasks against safety and efficiency baselines.

Dimension / 维度	Score	Comment / 评语
理论贡献 / Theoretical contribution	4/5	HOCBF 安全过滤 / High-order CBF safety filter
算法贡献 / Algorithmic contribution	5/5	Risk-aware ensemble distributional RL estimates policy uncertainty, then a high-order control barrier function acts as a safety filter.
实验贡献 / Experimental contribution	3/5	Simulation tests on multiple unsignalized-intersection tasks against safety and efficiency baselines.
工程贡献 / Engineering contribution	4/5	HOCBF safety filter dynamically adjusts constraints using joint uncertainty in unsignalized-intersection tasks.
博士课题价值 / PhD relevance	4/5	Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

HOCBF 安全过滤 / High-order CBF safety filter

$$u^* = \arg \min_u \|u - u_{RL}\|^2 \quad \text{s.t.} \quad L_f^m h(x) + L_g L_f^{m-1} h(x)u + \alpha(\psi_{m-1}(x)) \geq \rho(\sigma)$$

Symbol / 符号	含义	Meaning	Role / 建模作用
u_{RL}	强化学习建议动作	RL-proposed action	效率导向但未必安全
$h(x)$	安全函数	safety barrier function	$h \geq 0$ 表示安全集合
L_f, L_g	李导数	Lie derivatives	刻画动力学对安全函数的影响
σ	策略/状态不确定性	policy/state uncertainty	调节安全收紧程度
$\rho(\sigma)$	不确定性安全裕度	uncertainty-dependent safety margin	不确定性越高约束越保守

5.2 核心算法架构 / Core Algorithm Layout

先用风险感知 ensemble distributional RL 估计动作价值和不确定性，再通过 HOCBF 二次规划把 RL 动作投影到安全集合中。不确定性越大，屏障约束越保守。

An uncertainty-aware ensemble/distributional RL policy proposes actions, and a HOCBF quadratic program filters them into the safe set with uncertainty-dependent tightening.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

HOCBF 提供形式化安全条件，但效率取决于 RL 策略；若安全约束不可行，需要 fallback 或软约束设计。

The HOCBF layer can certify forward invariance under assumptions, but infeasible safety constraints require fallback handling.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 安全函数可正确表达交叉口危险集合。
- 车辆动力学和相对状态可观测。
- 不确定性估计与真实风险有足够相关性。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：普通 RL、安全 RL、固定 CBF、无不确定性调节的 HOCBF 和规则控制。

Validation / 验证: Simulation tests on multiple unsignalized-intersection tasks against safety and efficiency baselines.

Metrics / 指标: 碰撞率、成功率、平均时间、干预次数、不确定性校准和约束可行性。

Ablation / 消融: 必须比较无 HOCBF、固定 HOCBF、无 ensemble 不确定性、不同安全裕度函数。

Reviewer reading / 审稿式阅读提醒: 阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 8 figure references and 2 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 单车/局部自动驾驶视角多于集中式多车调度。
- HOCBF 对动力学模型准确性敏感。
- QP 频繁干预可能导致行为抖动或效率下降。

7.2 博士课题拓展点 / Research Extensions for PhD

- 把 JSSP scheduler 的动作作为 RL proposal，再用 HOCBF/MPC 投影到安全集合。
- 研究 CBF 约束不可行时的优先级回退和重调度。
- 用 conformal prediction 校准不确定性裕度，增强审稿可信度。

7.3 如何服务当前课题 / Use in This Project

Supports adding CBF/HOCBF safety filters or fallback logic to learned policies.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 安全裕度系数 / Safety margin coefficient, range [0.0, 4.0] .
系数越大, CBF 介入更早; 安全性提升但效率下降。
A larger margin triggers earlier intervention, improving safety but reducing efficiency.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator, 右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block, 紧跟符号表。
- 审稿人批注用 reviewer-note block, 博士拓展用 research-idea list。
- 交互参数用 `input[type=range]` + canvas 曲线, 打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.2	II. P RELIMINARIES
p.6	V. CONCLUSIONS

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 II. P RELIMINARIES 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.3 III. M ETHODOLOGIES 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.5 IV. I MPLEMENTATIONS 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.6 2 Bold: best performance. Policy update frequency $f = 10$ Hz. 作为局部论证段落阅读, 关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.6 V. CONCLUSIONS 总结贡献和未来方向，也常暴露作者承认的边界条件。/ Summarizes contributions and often reveals author-recognized boundaries.